

Homework #2. Due on Thursday, Sep 10th, 11:59pm on Canvas

Reading for this homework: Sections 1.4-1.6

Plan for next week (Sep 7,9): 1.6 (bases and dimension), 2.1 (linear transformations, kernels and images), maybe start 2.2 (matrix of a linear transformation).

In this homework you can use Theorems A and B below. They are both proved in 1.6 (Theorem B is stated slightly differently) and will also be proved in class on Mon, Sep 7.

Let V be a vector space over a field F . In class we defined V to be *finite-dimensional* if V has a finite spanning set. We then immediately proved that if V is finite-dimensional, then V has a finite basis (Lemma 3.5 from class).

Theorem A: Let V be a finite-dimensional vector space. Then any two bases of V have the same (finite) cardinality. This common cardinality is called the dimension of V .

Theorem B: Let V be a vector space of finite dimension n , and let S be a subset of V with $|S| = n$. Then S is a basis $\iff S$ is spanning $\iff S$ is linearly independent.

Problems:

Problem 1: Let V be a vector space over a field F .

- (a) Prove Lemma 3.4(1) from class: If S is a linearly independent subset of V which is not a basis, there exists a linearly independent subset T of V which strictly contains S (that is, $T \supseteq S$ and $T \neq S$). **Hint:** This can be proved similarly to one of the directions in Lemma 3.2 from class.
- (b) Assume that V is infinite-dimensional (that is, V is not finite-dimensional, as defined at the beginning of this homework). Use (a) to prove that V contains an infinite linearly independent subset.

Before the next problem we introduce the notion of characteristic of a field. Given a field F and $n \in \mathbb{N}$, we define $n_F \in F$ by $n_F = \underbrace{1_F + \dots + 1_F}_{n \text{ times}}$ (1_F added to itself n times). If $n_F \neq 0_F$ for all $n \in \mathbb{N}$, one defines the characteristic of F (denoted $\text{char}(F)$) to be 0; otherwise, $\text{char}(F)$ is the smallest n such that $n_F = 0_F$. For example, $\text{char}(\mathbb{Q}) = \text{char}(\mathbb{R}) = \text{char}(\mathbb{C}) =$

0 while $\text{char}(\mathbb{Z}_p) = p$ for any prime p . It is not difficult to show that $\text{char}(F)$ is either 0 or a prime number for any field F .

Problem 2: Let F be a field, V a vector space over F and $u, v \in V$ distinct elements such that $\{u, v\}$ is linearly independent.

- (a) Prove that $u + v \neq v$ and $\{u + v, v\}$ is linearly independent.
- (b) Assume that $\text{char}(F) \neq 2$. Prove that $u + v \neq u - v$ and $\{u + v, u - v\}$ is linearly independent.

Note that if $\text{char}(F) = 2$, then $v + v = 0_V$ for any v (since $v + v = 1_F \cdot v + 1_F \cdot v = (1_F + 1_F) \cdot v = 2_F \cdot v = 0_F \cdot v = 0_V$), so $-v = v$ and hence $u - v = u + v$.

Problem 3: Let F be a field, $n \in \mathbb{N}$ and $\mathcal{P}_n(F)$ the vector space of all polynomials of degree $\leq n$ with coefficients in F . Let S be a subset of $\mathcal{P}_n(F)$ such that $0 \notin S$ and any two elements of S have distinct degrees. Prove that S is linearly independent. There is a minor hint at the end of the assignment.

Problem 4:

- (a) Let p be a prime, $n \geq 1$ an integer, and let $V = \mathbb{Z}_p^n$, the standard n -dimensional vector space over $\mathbb{Z}_p$ (integers mod p). How many elements does V have?
- (b) Find all ordered bases for $\mathbb{Z}_2^2$. How many unordered bases does it have? (An unordered basis is a basis in the usual sense; an ordered basis is a basis with a chosen order on its elements).
- (c) Now let p be any prime. How many ordered bases does $\mathbb{Z}_p^2$ have?

Hint for (b) and (c): Use Problem 8 from HW#1 and one of the results stated at the beginning of this assignment. See the end of the assignment for an additional hint for (c).

Problem 5: Let F be a field, X a set and $V = \text{Func}(X, F)$, the set of all functions from X to F . As discussed in Lecture 2, V has a natural structure of a vector space over F . For each $x \in X$ define the function $\delta_x \in V$ by $\delta_x(y) = 0$ for all $y \neq x$ and $\delta_x(x) = 1$. Let $S = \{\delta_x : x \in X\}$.

- (a) Prove that the set S is always linearly independent.
- (b) Assume that S is finite. Prove that S is spanning for V (and hence by (a) is a basis for V).
- (c) Now prove that if S is infinite, then S is not spanning for V . **Hint:** you can find a simple explicit function which does not lie in $\text{Span}(V)$.

Problem 6: Let F be a field and $V = \text{Mat}_n(F)$ the vector space of all $n \times n$ matrices over F . For each $1 \leq i, j \leq n$ let $e_{ij} \in \text{Mat}_n(F)$ denote the matrix whose (i, j) -entry is equal to 1 and all other entries are equal to 0.

- (a) Prove that the set $B = \{e_{ij} : 1 \leq i, j \leq n\}$ is a basis of V .
- (b) Let U be the set of upper-triangular matrices in V , that is, U consists of all matrices whose (i, j) -entry is equal to 0 for all $i > j$. Prove that U is a subspace and find a basis of U and $\dim(U)$. **Hint:** U has a basis which is a subset of B .
- (c) Now find a basis for $W = \mathfrak{sl}_n(F)$, the set of all matrices in V with zero trace (recall that W was shown to be a subspace in HW#1.2). **Hint:** W does not have a basis which is a subset of V , but you may want to look for a basis consisting of matrices with as few nonzero entries as possible.

Problem 7: This is a continuation of Problem 1 from HW#1. This time we are looking at $\mathbb{Q}$ -subspaces of $\mathbb{C}$. A complex number $\alpha \in \mathbb{C}$ is called *algebraic* if α is a root of a nonzero polynomial with rational coefficients, that is, if there exist rational numbers $q_0, \dots, q_n$, not all 0, such that $\sum_{i=0}^n q_i \alpha^i = 0$. A complex number is called *transcendental* if it is not algebraic (for example, one can show that e and π are transcendental).

Suppose you are given a specific transcendental number $\alpha \in \mathbb{C}$. Use it to construct, for every $n \in \mathbb{N}$, an explicit $\mathbb{Q}$ -subspace of $\mathbb{C}$ of dimension exactly n .

Hint for 3: Start with a concrete example, say, $F = \mathbb{R}$, $n = 2$, and S consisting of 3 polynomials of degrees 0, 1 and 2, respectively, and look for a simple reason why S is linearly independent. Then try to generalize that argument to an arbitrary S satisfying the given conditions.

Hint for 4(c): How many choices are there to choose the first element of an ordered basis. Once the first element has been chosen, how many choices are there for the second element?